Phoenix Robotics Engineering Presentation Draft

Has to be under 15 mins ~

Colin: Hello, I am Colin Price and I am the CEO of Phoenix Robotics, a team dedicated to soaring to new heights through cutting edge technology. This is my 2nd year on the team and I would like to start off with our number one priority, Safety.

Josh: Hello, My name is Joshua Chao, this is my second year as a member of phoenix robotics. I'm the chief safety officer, and I'm OSHA 10 certified-and 3 other members on the team are OSHA-10 certified as well. I ensure proper protocols, risk management behaviors and our company JSA is known by all company members. I ensured that the MATE 25 amp fuse was properly implemented into our electronics and that the emergency off switch functions. I make sure that everyone is wearing proper PPE, tying long hair back, and wearing closed-toed shoes. The ROV has no sharp edges as they were sanded down to prevent harm to company members. The thrusters are all internal to further reduce chance of injury and they are shrouded to the IP20 standard. Acrylic sheets above the camera and control boxes protect electronics from water damage and shrapnel.

Carson: (rov design) , now let's talk about the tasks that the rov it completes

Cecilia: (Describes task and then missions), our ROV has many specialized subsystems tod complete this missions/task

Chara: (Talk about subsystems, how they help complete tasks.)

- AT9O, first year on team
- Tasks hard this year
- Manipulators fragile but have been improved on throughout this year
-

Hello, my name is Xia Collier and my role on this team is the assistant technician officer. The tasks in this year's competition were exceptionally specialized and our team had to come up with creative solutions to them. The manipulators we applied allow us to accurately move light-weight objects accompanied by our buoyancy balloons that allow us to gracefully move vertically without causing harm or disruption to the environment or company equipment. Second, our hook allows us to carry heavy objects without risking damage to our carefully crafted manipulators. Another unique feature of our ROV is the LED light utilized to cure coral and our vertical profiler that is used for task 3.0.

Mohammed: (Talk about vp what it does for the environment, the code, and how it works.) , I'll let Gavin tell you how he built these environments.
Hello, my name is Mohammed Badri and my role on this team is the Vertical Profile Specialist. The vertical profiler is 1.7 feet with a 3-inch diameter inside there is an Arduino Uno with 2 cytrons, 2 9v batteries with a 7.5 amp fuse, 2 air pumps with a 2.5 LPM flow rate, and an internal bladder stores air initially, and with 2 air pumps transferring pneumatic power the air valve makes sure it does not leak. Additionally, the Wifi transmitter has a NodeMCU connected to an RTC module e. When we first turn the

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Arduino on using the button on the battery case and that initiates a 90-second delay after the delay it sucks any remaining air from the orange balloon and puts it in the blue then because of the buoyancy change the vertical profiler goes down for 30 seconds and then the orange balloon sucks air for 16 seconds then it has another 30-second delay at which point the wifi transmitter communicates with my laptop sending a signal every second. I'll let Gavin tell you how he built environments for the ROV.

Gavin: (Talk about the environments and what he learned from being on the team),

Hello my name is Gavin Udagawa this is my first year being a part of Phoenix Robotics. My role within the company is the prop specialist, this means that I make and maintain the props for the ROV. My role is essential because I help to create a realistic environment for the ROV, which allows for effective practice. During these practices I work poolside and would take and give the ROV any props that it need, to ensure a more realistic training experience. An example of this is with task 2.5 where I constructed the fish release sites to be an accurate simulation of the actual thing, along with giving the ROV the fish container to effectively simulate the process of it. My time on this team has taught me how to both communicate and coordinate with people in an effective manner as to work towards a shared goal.

Colin: This year our team has had an immense amount of self-teaching, as half the members are new this year. The team has largely focused on passing down skills to new team members, such as soldering, coding, CAD, and team skills while constructing the ROV.

Kris: (talk about what you learn segue into build vs buy vs used),

Colin: Our team has put a large effort into outreach, most exemplified by our two weeks spent at the Florida State fair educating children and adults on ROV and the MATE ROV competition. We even set up a mini rov exhibit and allowed kids to get a simplified idea on how ROVs work and how to control them. We mentored new teams (specific team names.) often joining zoom calls with them and helping them solve issues they had well building their own ROVs.

Carson: (Ending remarks and acknowledgements - begin questioning)

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Talk aboyut reedundancy
Talk about waterproofing of tube
Find someone to talk about UN goads
Have kris not tlk about mentors
Find a way to sell product to our customer which feeds our employees with skills for future careers
Carson talk about claws
Austin talk about ballast and calculation of flinking
Gavin talk about vertical profiler buoyancy engine\
Mohammed talk about the sensor for the vertical profiler
Kris talk about cando and amroc and the partnership
Cfarson talk about design talk about interests
Know about systems
Josh talk about cameras
Bryce talk about tube and waterproofing of tube and tray
Dont care about the rubric
Make sure you really know what you are talking about and who would best know anything if a question is directed toward you if you dont know how to answer